

Motor Control Board

Variant: DRAFT

2026-09-03

Rev + (Unreleased)

Page	Index	Page	Index	Page	Index	Page	Index
1	Cover Page	11	Motor Control 4	21	Misc - Board Version, Switch	31	
2	Block Diagram	12	Motor Control 5	22	Power - Sequencing	32	
3	Project Architecture	13	Motor Control 6	23	Revision History	33	
4	Power - Connectors	14	Interface - USB 2.0	24		34	
5	Power - Generation	15	Interface - FD-CAN	25		35	
6	MCU - Power	16	Interface - Interconnects	26		36	
7	MCU - IOs	17	User - LED Indicators & Buzzer	27		37	
8	Motor Control 1	18	Sensing - Temperature & Humidity	28		38	
9	Motor Control 2	19	Sensing - Battery	29		39	
10	Motor Control 3	20	Misc - Holes, Fiducials	30		40	

TOP VIEW

kibot_Mimage_png_3d_viewer_angled_top

BOTTOM VIEW

kibot_Mimage_png_3d_viewer_angled_bottom

NOTES

Comment

Not fitted components are marked as **X**

DRAFT - Very early stage of schematic, ignore details.

PRELIMINARY - Close to final schematic.

CHECKED - There shouldn't be any mistakes. Contact the engineer if you find any.

RELEASED - A board with this schematic has been sent to production.

Block Diagram

File: Block Diagram.kicad_sch

Revision History

File: Revision History.kicad_sch

Project Architecture

File: Project Architecture.kicad_sch

Power - Sequencing

File: Power - Sequencing.kicad_sch

DESIGN CONSIDERATIONS

DESIGN NOTE:
Example text for
informational design
notes.

DESIGN NOTE:
Example text for
debug notes.

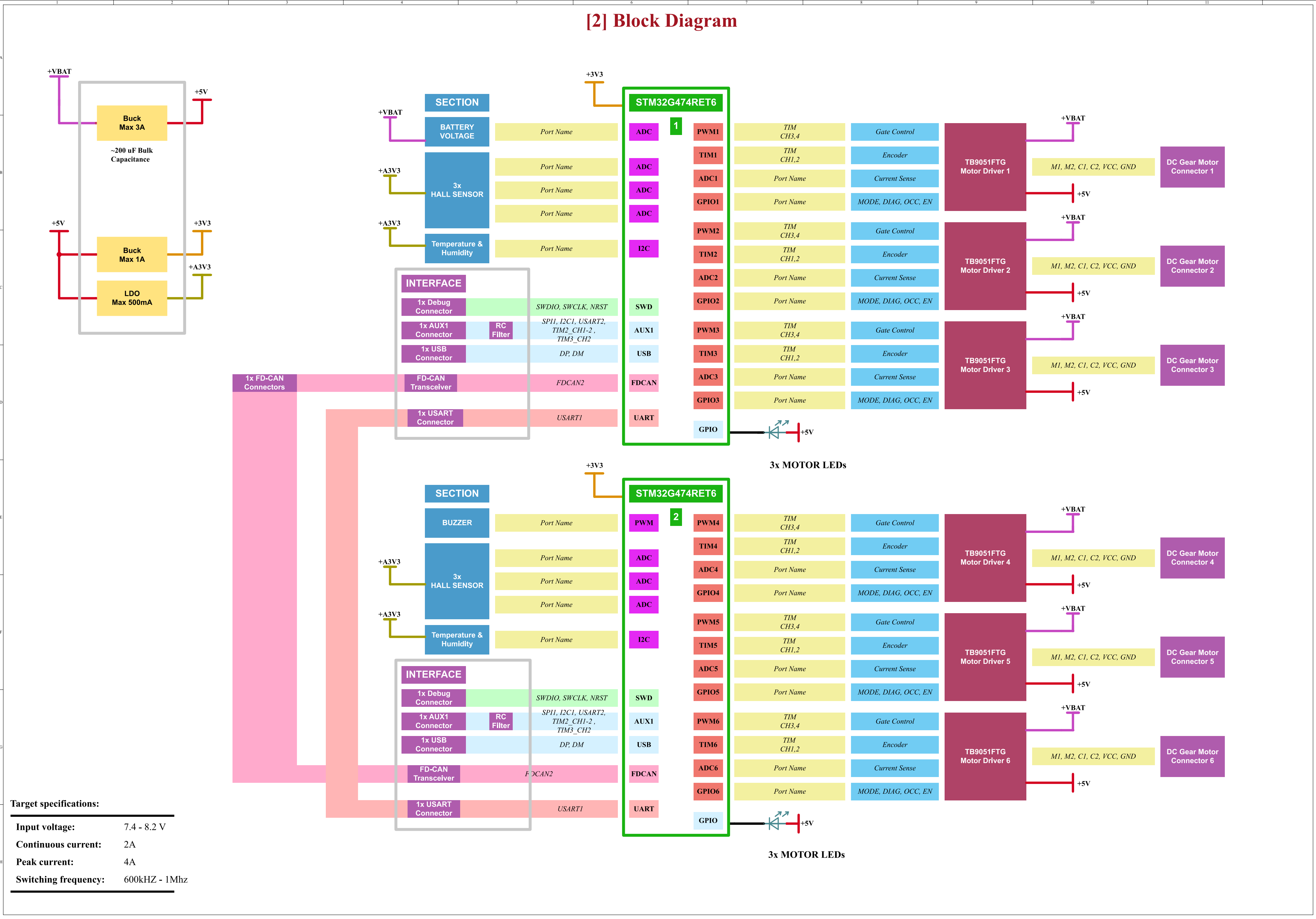
DESIGN NOTE:
Example text for
cautionary design
notes.

DESIGN NOTE:
Example text for
critical design
notes.

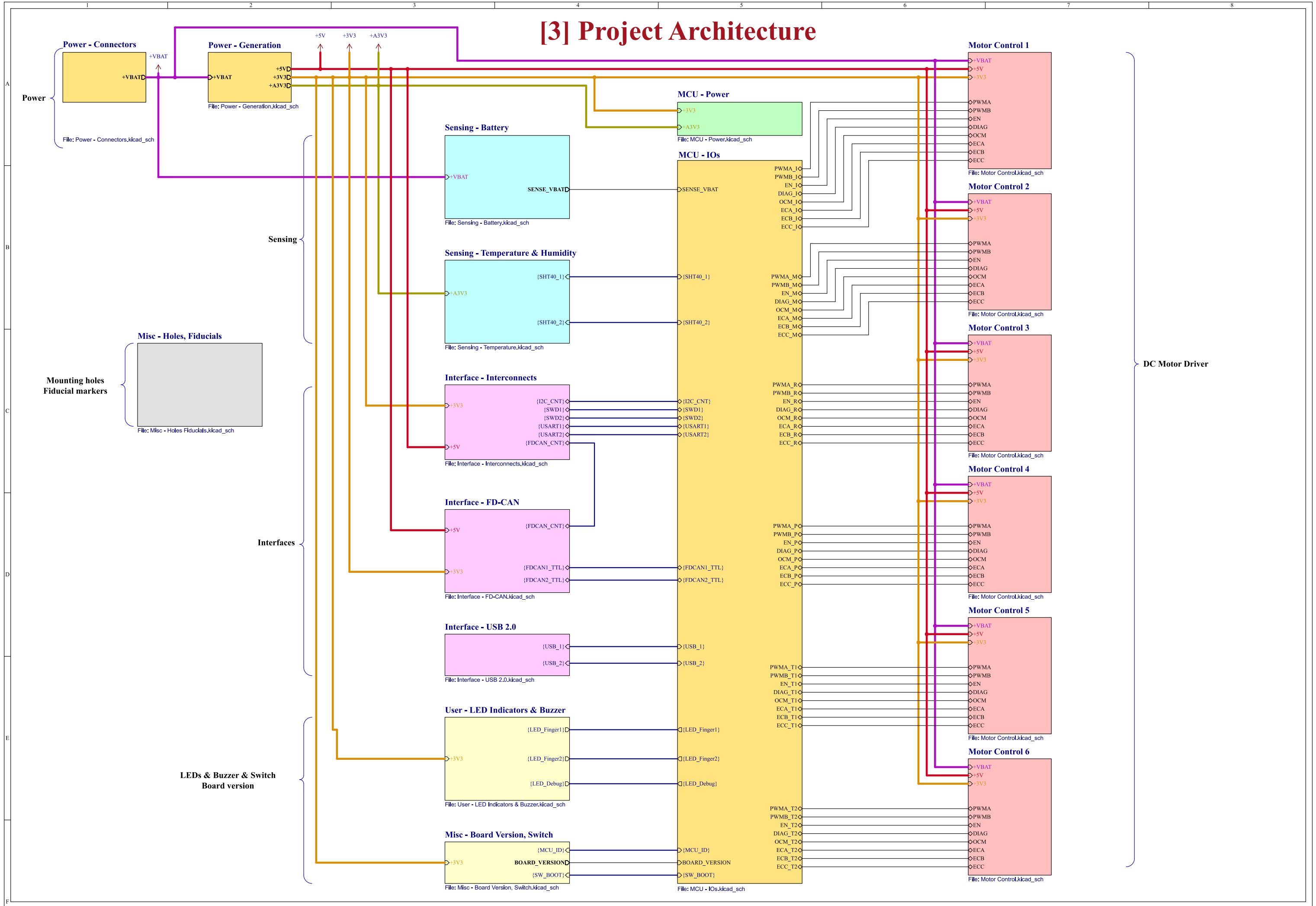
LAYOUT NOTE:
Example text for
critical layout
guidelines.

Date: 03-Sep-2026

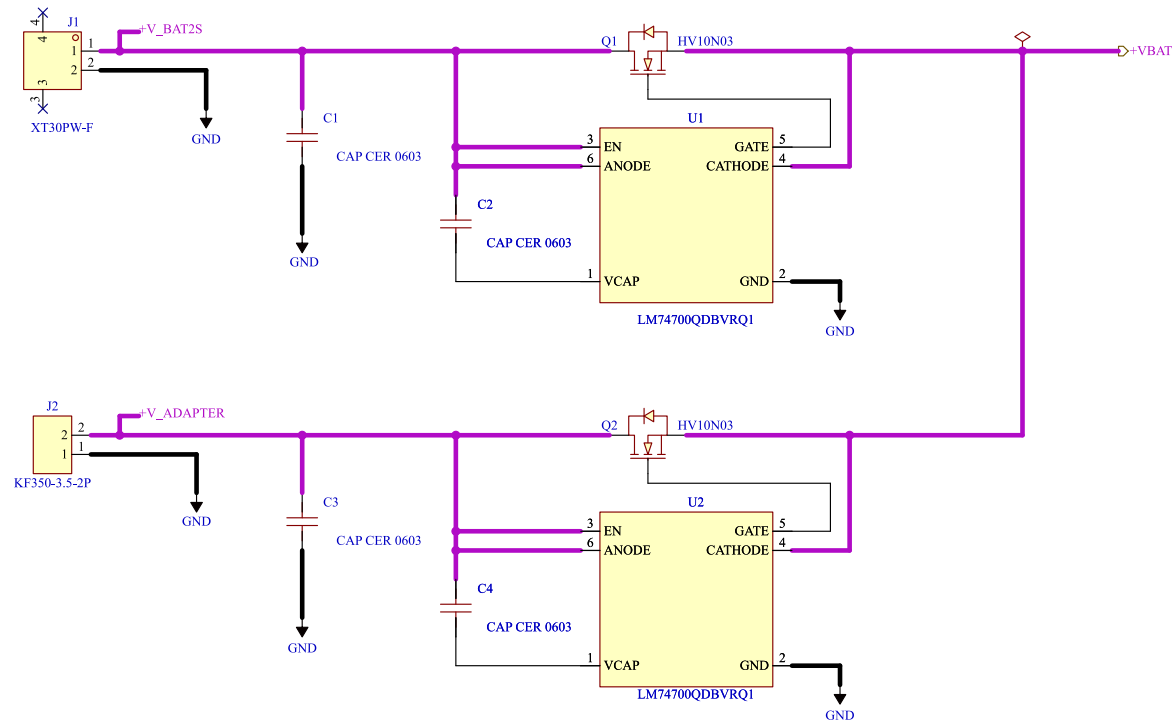
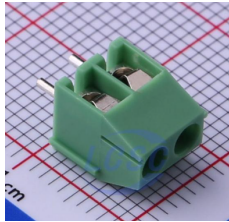
[2] Block Diagram



[3] Project Architecture

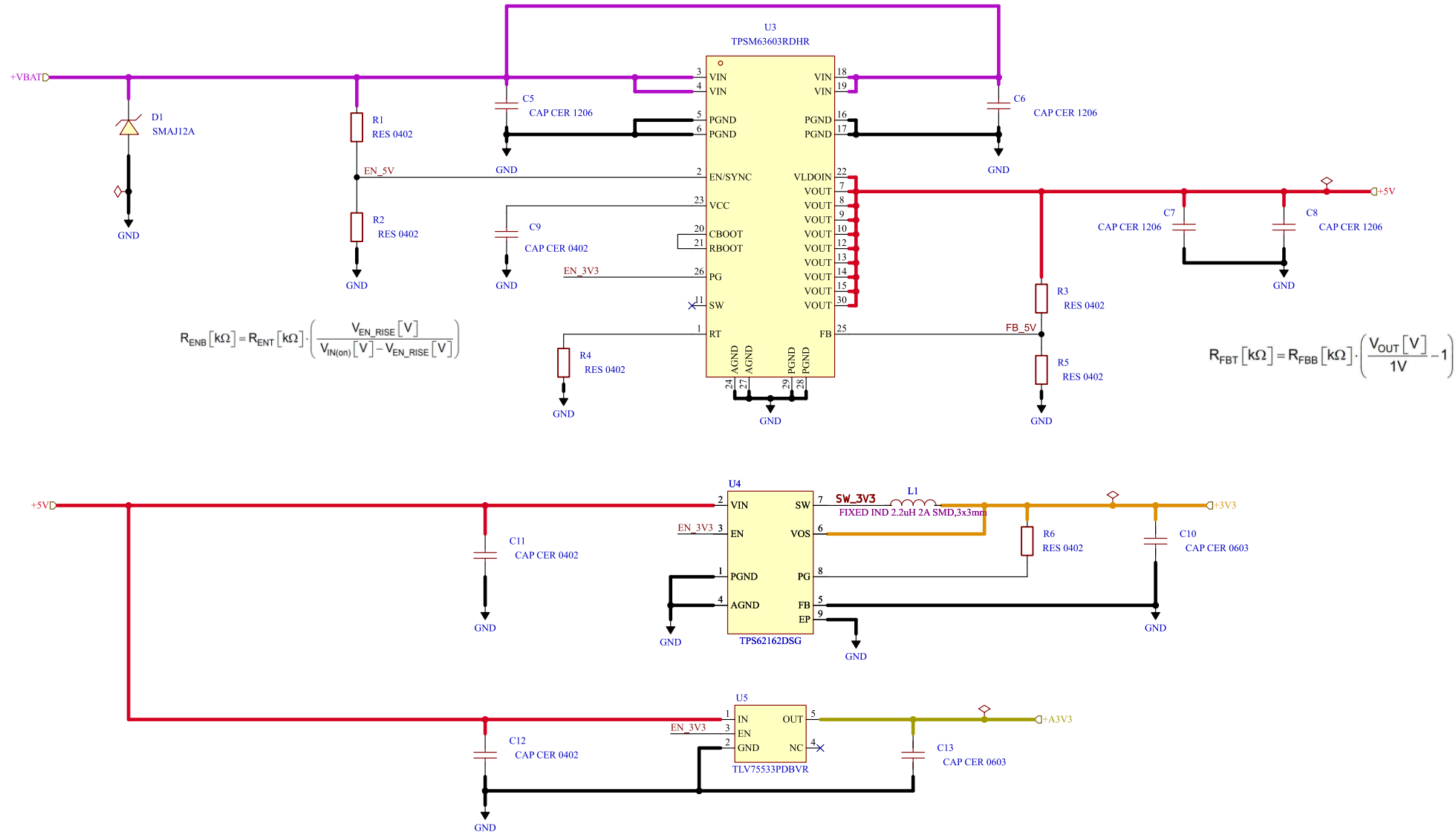


[4] Power - Connectors



Power Connectors - Path Control

[5] Power - Generation

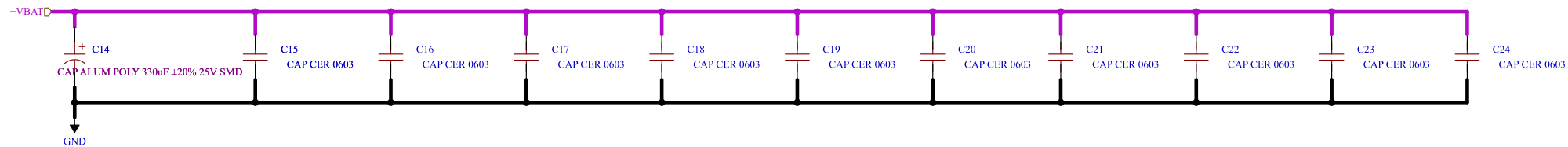


DESIGN NOTE:
PI filter with inductor would give better EMC performance but has higher bulk capacitance requirements. Ferrite is a good compromise.

LAYOUT NOTE:
Route feedback traces away from switching node.

LAYOUT NOTE:
Minimize switching node loop areas.

Voltage Regulators



DC BULK

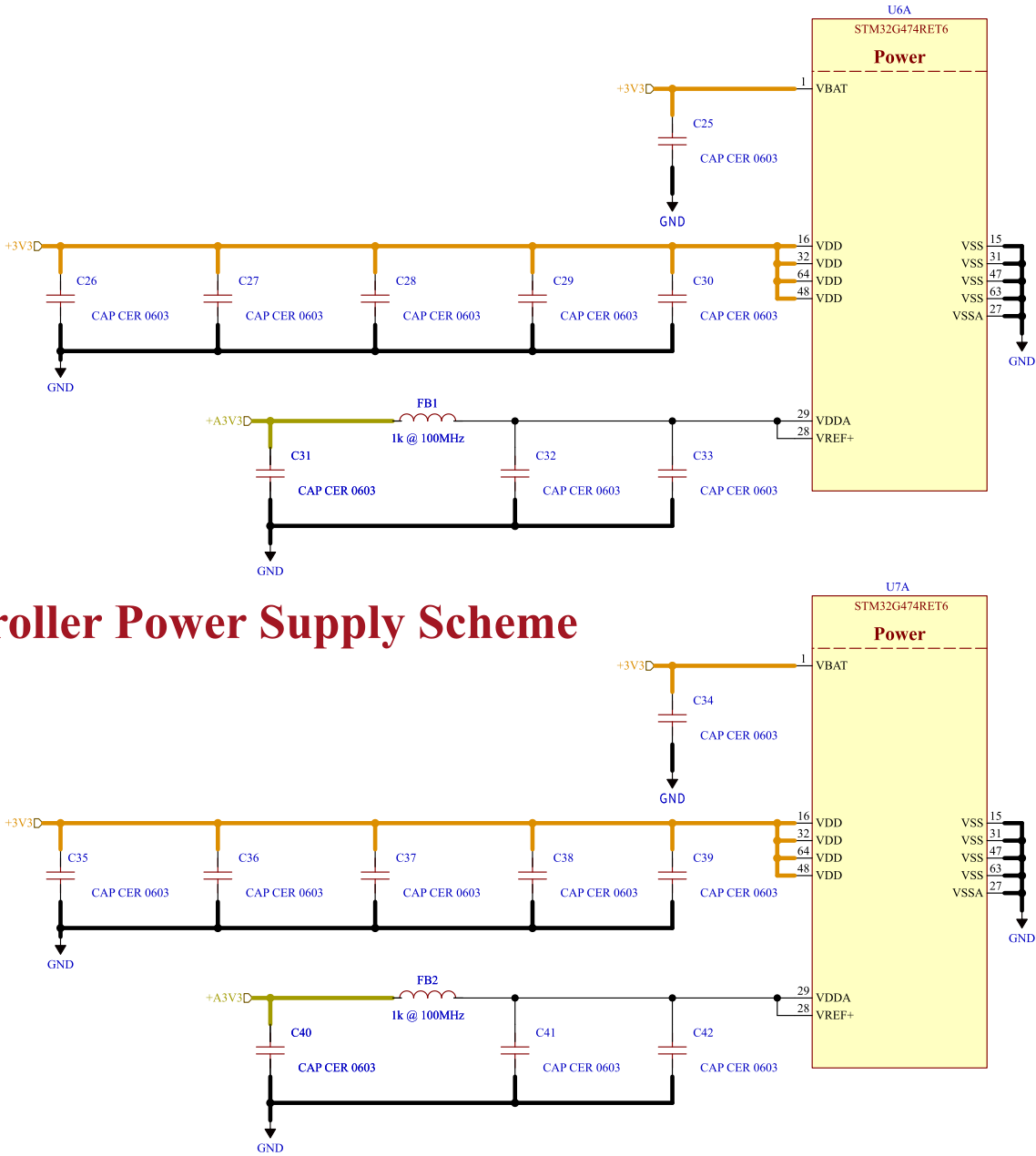
DC-Link Bulk Capacitors. 330uF + 10 x 10 uF = 430uF (35V)

Test Points

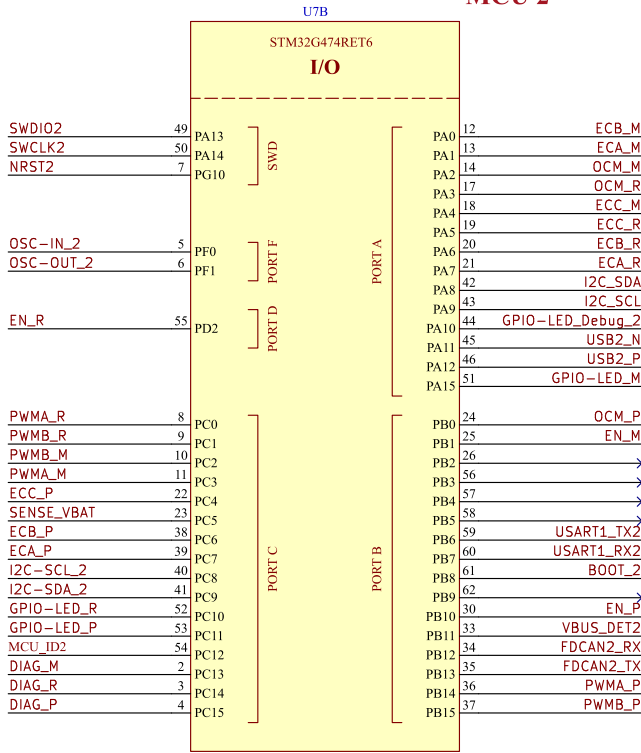
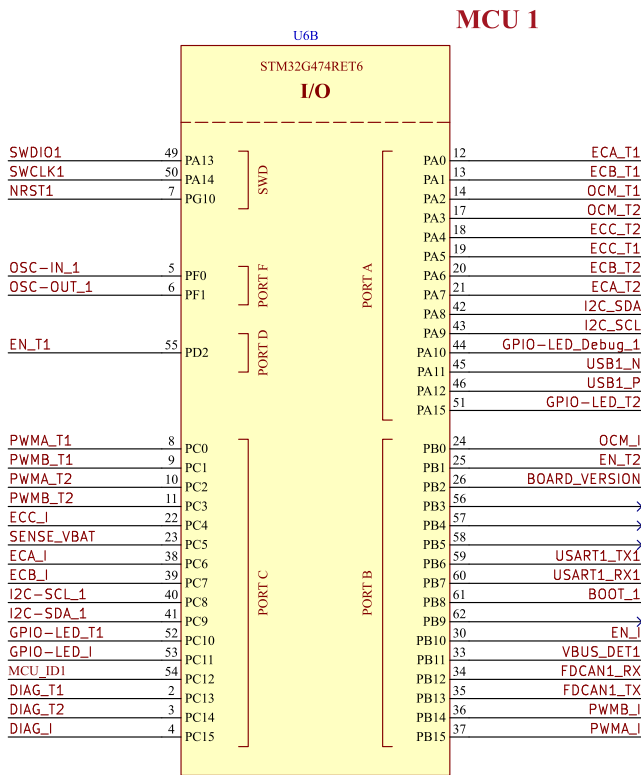


Microcontroller Power Supply Scheme

LAYOUT NOTE:
Place each capacitor close
to the relevant pin.
Minimize loop inductance.



[7] MCU - IOs



DESIGN NOTE:
We use the LQFP64 version to have extra pins for controlling an external fan.

DESIGN NOTE:
Do not forget to flash option bytes when flashing for the first time.

LAYOUT NOTE:
Footprint deviates from datasheet and IPC7351 to allow for soldermask sliver between pads according to PCBWay design rules.

SENSE_VBAT → SENSE_VBAT

Battery Voltage Sense

BOARD_VERSION → BOARD_VERSION

Board Version Selection

MCB to MB - USART

USART1_RX1 → USART1
USART1_TX1 → USART1

USART1_RX2 → USART2
USART1_TX2 → USART2

SWDIO1 → SWD1
SWCLK1 → SWD1
NRST1 → SWD1

SWDIO2 → SWD2
SWCLK2 → SWD2
NRST2 → SWD2

SWD (Debug) Port

I2C_SDA → I2C_CNT
I2C_SCL → I2C_CNT

MCB to MB - I2C

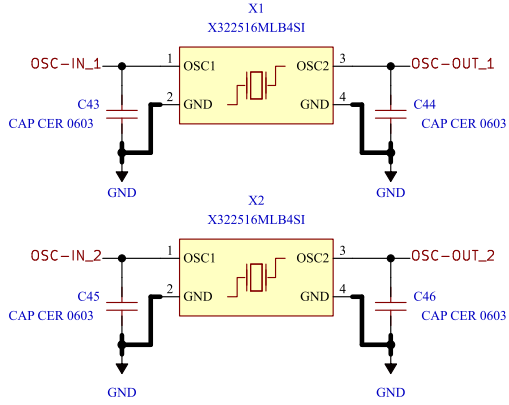
FDCAN1_RX → FDCAN1_TTL
FDCAN1_TX → FDCAN1_TTL

FDCAN2_RX → FDCAN2_TTL
FDCAN2_TX → FDCAN2_TTL

5Mbps FD-CAN Bus

MCU_ID1 → MCU_ID
MCU_ID2 → MCU_ID

MCU ID



Oscillator HSE

GPIO-LED_I → LED_Finger1
GPIO-LED_T1 → LED_Finger2
GPIO-LED_T2 → LED_Finger2

GPIO-LED_P → LED_Debug
GPIO-LED_R → LED_Debug
GPIO-LED_M → LED_Debug

GPIO-LED_Debug_1 → LED_Debug
GPIO-LED_Debug_2 → LED_Debug

LEDs

I2C-SDA_1 → SHT40_1
I2C-SCL_1 → SHT40_1

I2C-SDA_2 → SHT40_2
I2C-SCL_2 → SHT40_2

Temperature and Humidity

BOOT_1 → SW_BOOT
BOOT_2 → SW_BOOT

SWITCH

USB1_P → USB_1
USB1_N → USB_1
VBUS_DET1 → USB_1

USB2_P → USB_2
USB2_N → USB_2
VBUS_DET2 → USB_2

USB PORT

PWMA_I → PWMA_I
PWMB_I → PWMB_I
EN_I → EN_I
DIAG_I → DIAG_I
OCM_I → OCM_I
ECA_I → ECA_I
ECB_I → ECB_I
ECC_I → ECC_I

PWMA_M → PWMA_M
PWMB_M → PWMB_M
EN_M → EN_M
DIAG_M → DIAG_M
OCM_M → OCM_M
ECA_M → ECA_M
ECB_M → ECB_M
ECC_M → ECC_M

PWMA_R → PWMA_R
PWMB_R → PWMB_R
EN_R → EN_R
DIAG_R → DIAG_R
OCM_R → OCM_R
ECA_R → ECA_R
ECB_R → ECB_R
ECC_R → ECC_R

PWMA_P → PWMA_P
PWMB_P → PWMB_P
EN_P → EN_P
DIAG_P → DIAG_P
OCM_P → OCM_P
ECA_P → ECA_P
ECB_P → ECB_P
ECC_P → ECC_P

PWMA_T1 → PWMA_T1
PWMB_T1 → PWMB_T1
EN_T1 → EN_T1
DIAG_T1 → DIAG_T1
OCM_T1 → OCM_T1
ECA_T1 → ECA_T1
ECB_T1 → ECB_T1
ECC_T1 → ECC_T1

PWMA_T2 → PWMA_T2
PWMB_T2 → PWMB_T2
EN_T2 → EN_T2
DIAG_T2 → DIAG_T2
OCM_T2 → OCM_T2
ECA_T2 → ECA_T2
ECB_T2 → ECB_T2
ECC_T2 → ECC_T2

Motor Driver

Battery Voltage Sense

SENSE_VBAT → TP8

Board Version Selection

BOARD_VERSION → TP12

SWD (Debug) Port

NRST2 → TP16
SWCLK2 → TP18
SWDIO2 → TP20

5Mbps FD-CAN Bus

FDCAN1_TX → TP28
FDCAN1_RX → TP30
FDCAN2_TX → TP33
FDCAN2_RX → TP35

LEDs

GPIO-LED_I → TP37
GPIO-LED_T1 → TP39
GPIO-LED_T2 → TP41
GPIO-LED_P → TP43
GPIO-LED_R → TP45
GPIO-LED_M → TP47
GPIO-LED_Debug_1 → TP49
GPIO-LED_Debug_2 → TP51

Temperature and Humidity

I2C-SDA_1 → TP53
I2C-SCL_1 → TP55
I2C-SDA_2 → TP57
I2C-SCL_2 → TP59

USB PORT

USB1_P → TP63
USB1_N → TP65
VBUS_DET1 → TP66

MCB to MB - I2C

I2C_SCL → TP76
I2C_SDA → TP78

Buzzer

MCU_ID1 → TP84
MCU_ID2 → TP86

MCU ID

MD Index

PWMA_I → TP9
PWMB_I → TP10
EN_I → TP11
DIAG_I → TP13
OCM_I → TP14
ECA_I → TP15
ECB_I → TP17
ECC_I → TP19

MD Middle

PWMA_M → TP22
PWMB_M → TP24
EN_M → TP26
DIAG_M → TP27
OCM_M → TP29
ECA_M → TP31
ECB_M → TP32
ECC_M → TP34

MD Ring

PWMA_R → TP36
PWMB_R → TP38
EN_R → TP40
DIAG_R → TP42
OCM_R → TP44
ECA_R → TP46
ECB_R → TP48
ECC_R → TP50

MD Pinky

PWMA_P → TP52
PWMB_P → TP54
EN_P → TP56
DIAG_P → TP58
OCM_P → TP60
ECA_P → TP61
ECB_P → TP62
ECC_P → TP64

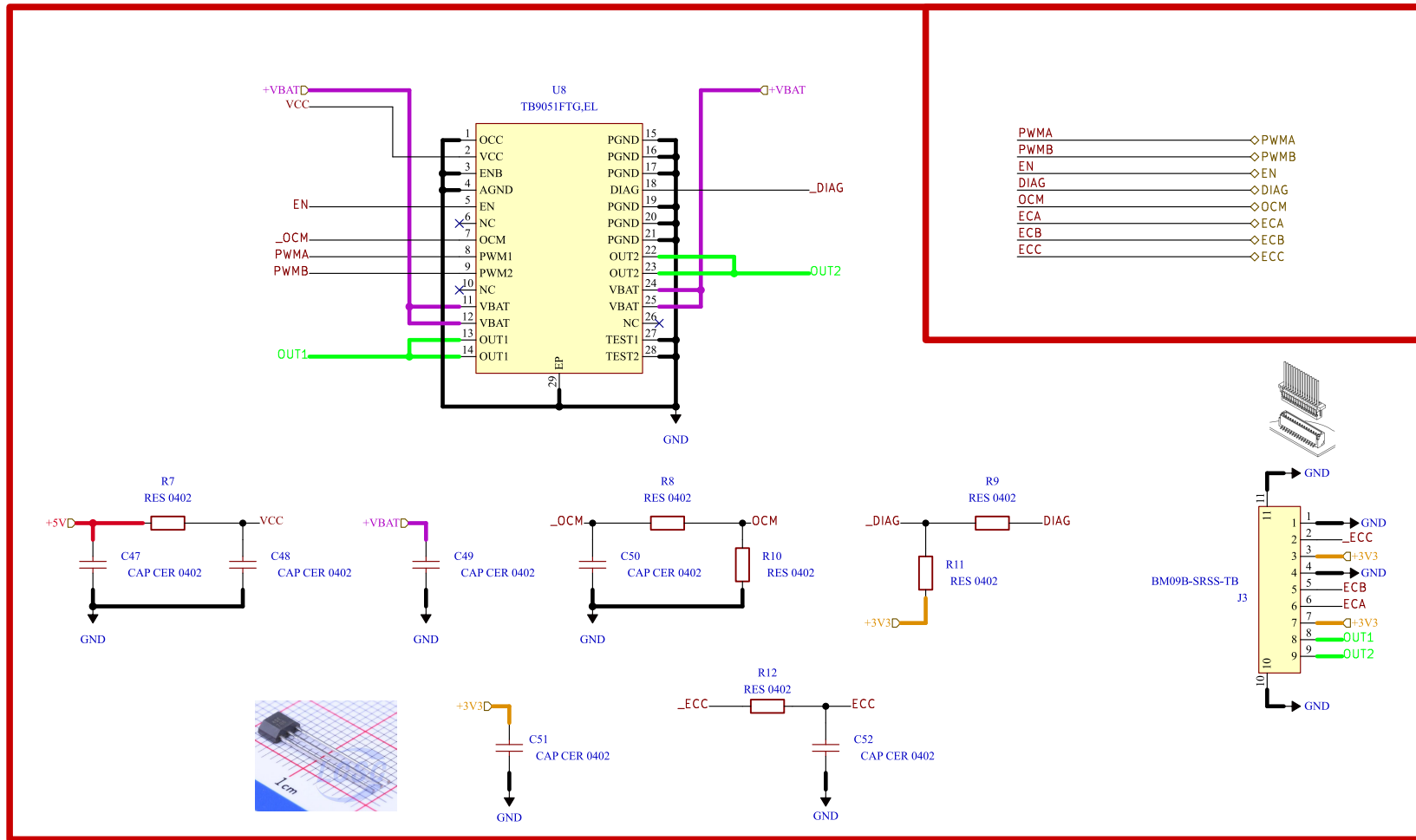
MD Thumb 1

PWMA_T1 → TP67
PWMB_T1 → TP69
EN_T1 → TP71
DIAG_T1 → TP73
OCM_T1 → TP74
ECA_T1 → TP75
ECB_T1 → TP77
ECC_T1 → TP79

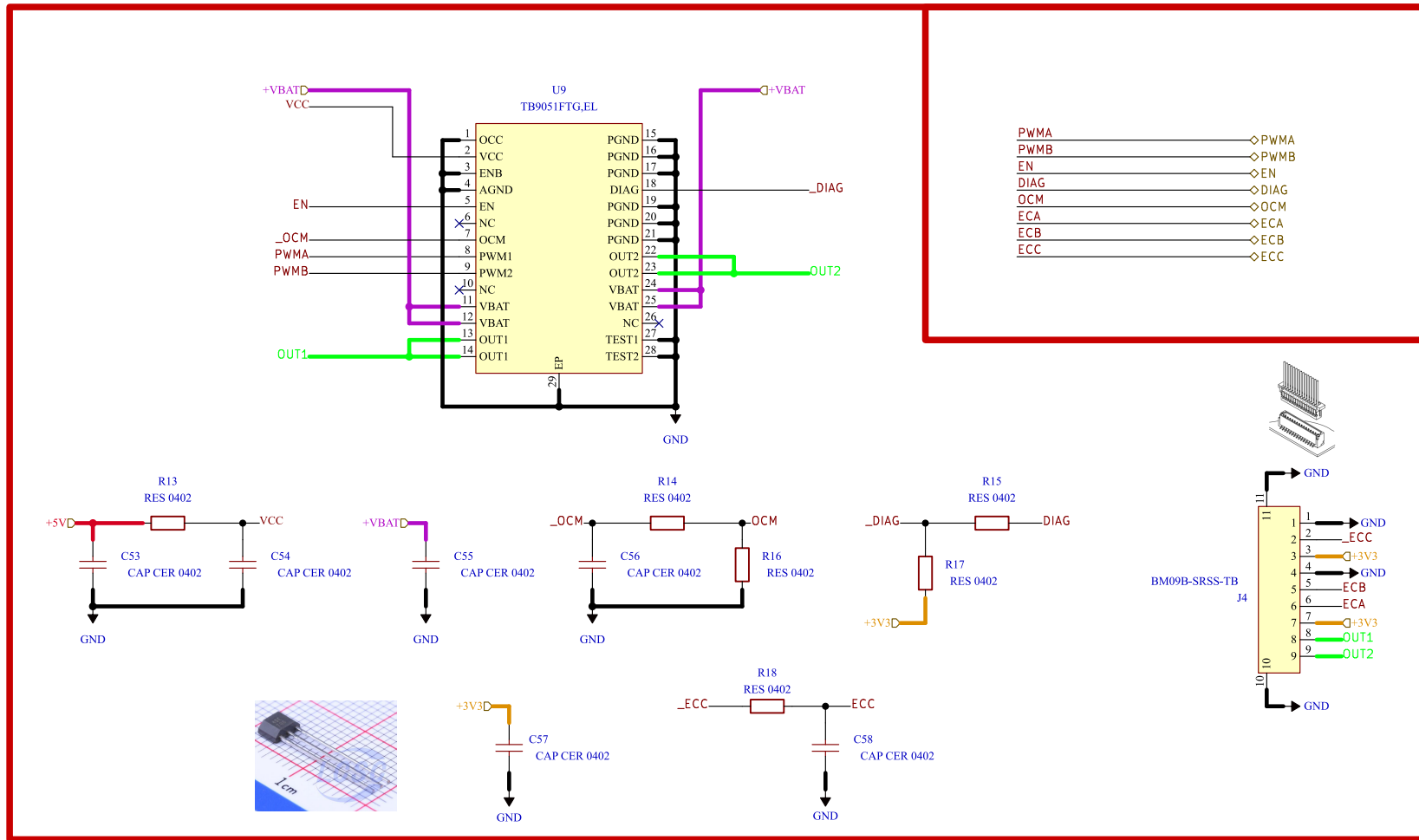
MD Thumb 2

PWMA_T2 → TP80
PWMB_T2 → TP81
EN_T2 → TP82
DIAG_T2 → TP83
OCM_T2 → TP85
ECA_T2 → TP87
ECB_T2 → TP88
ECC_T2 → TP89

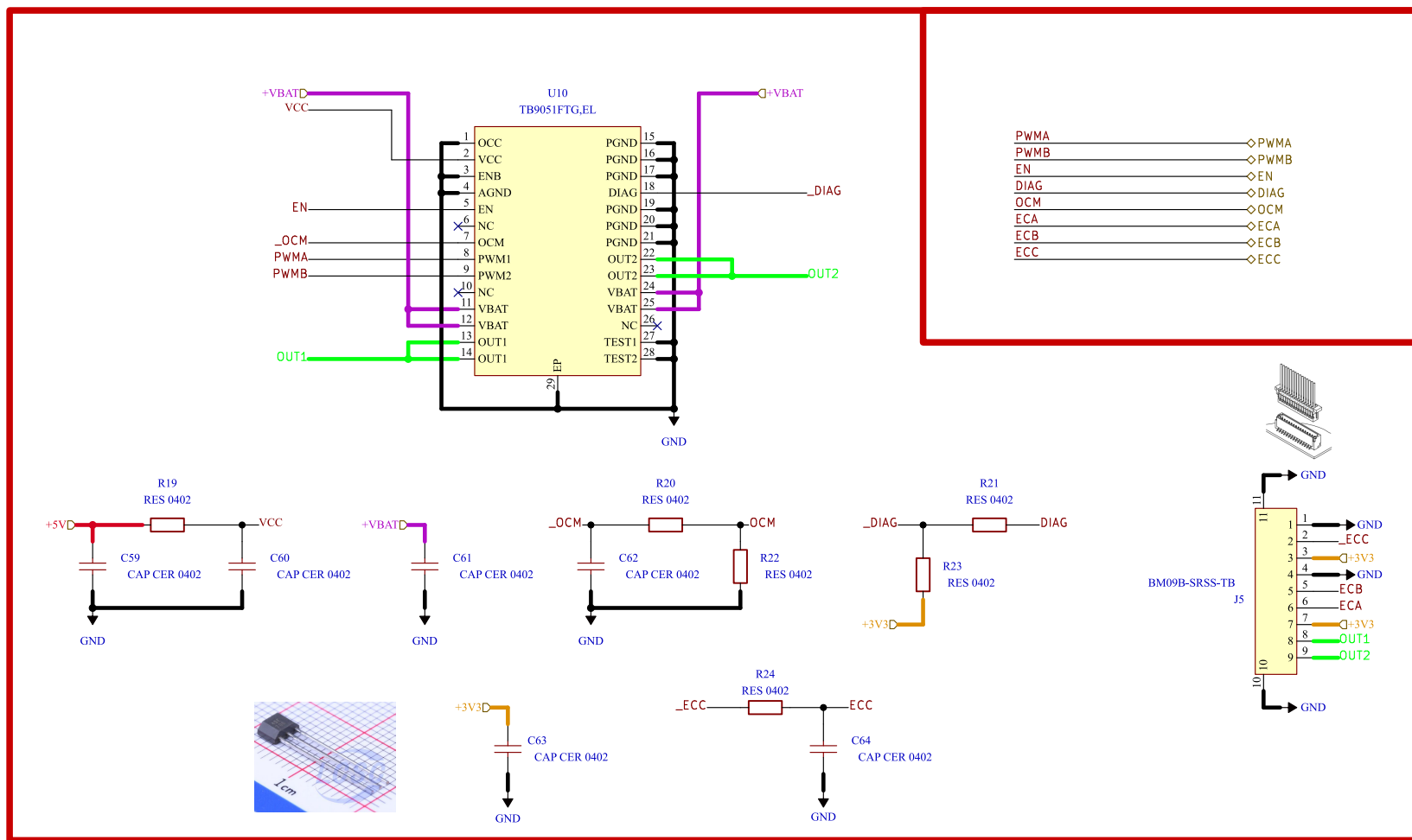
[8] Motor Control 1



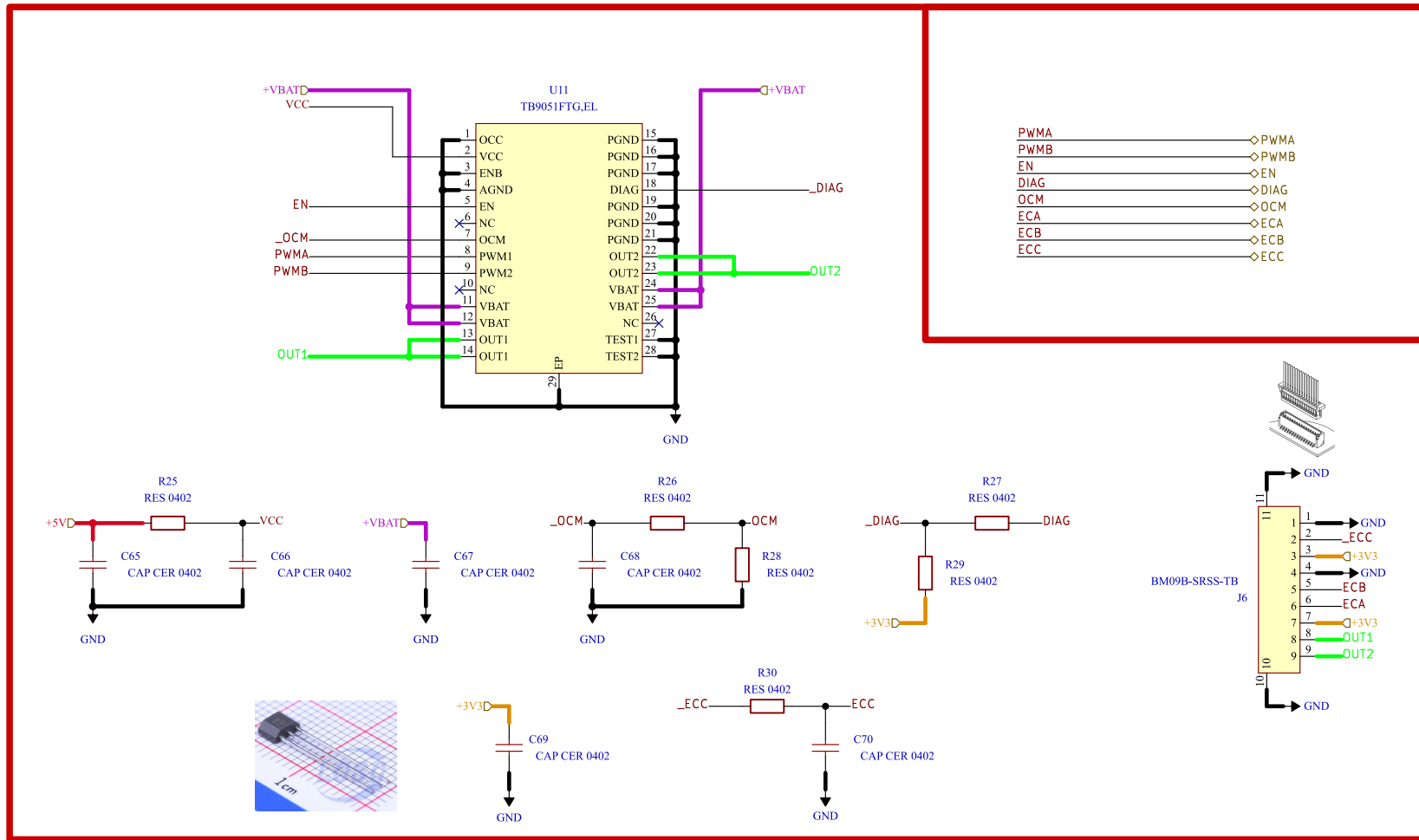
[9] Motor Control 2



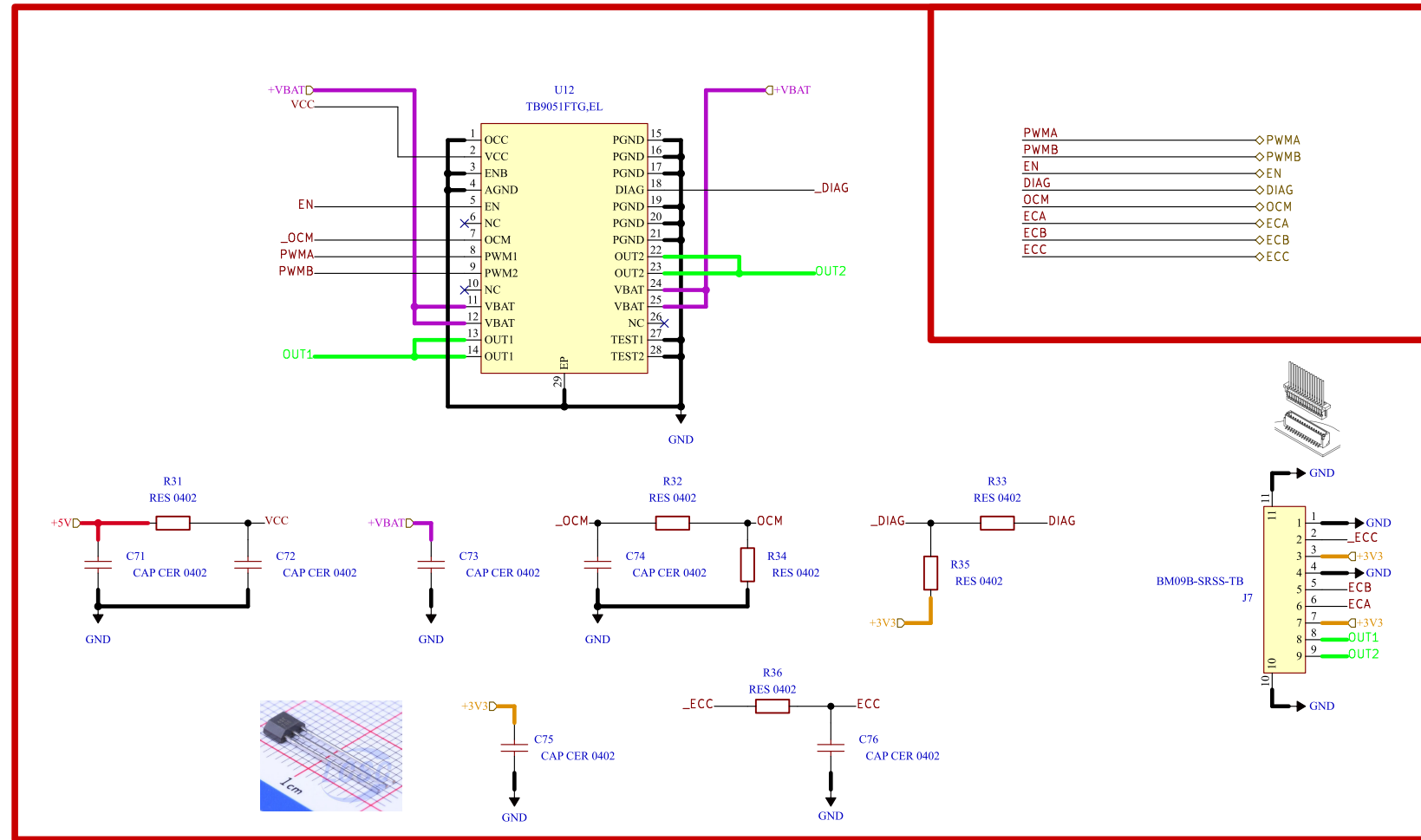
[10] Motor Control 3



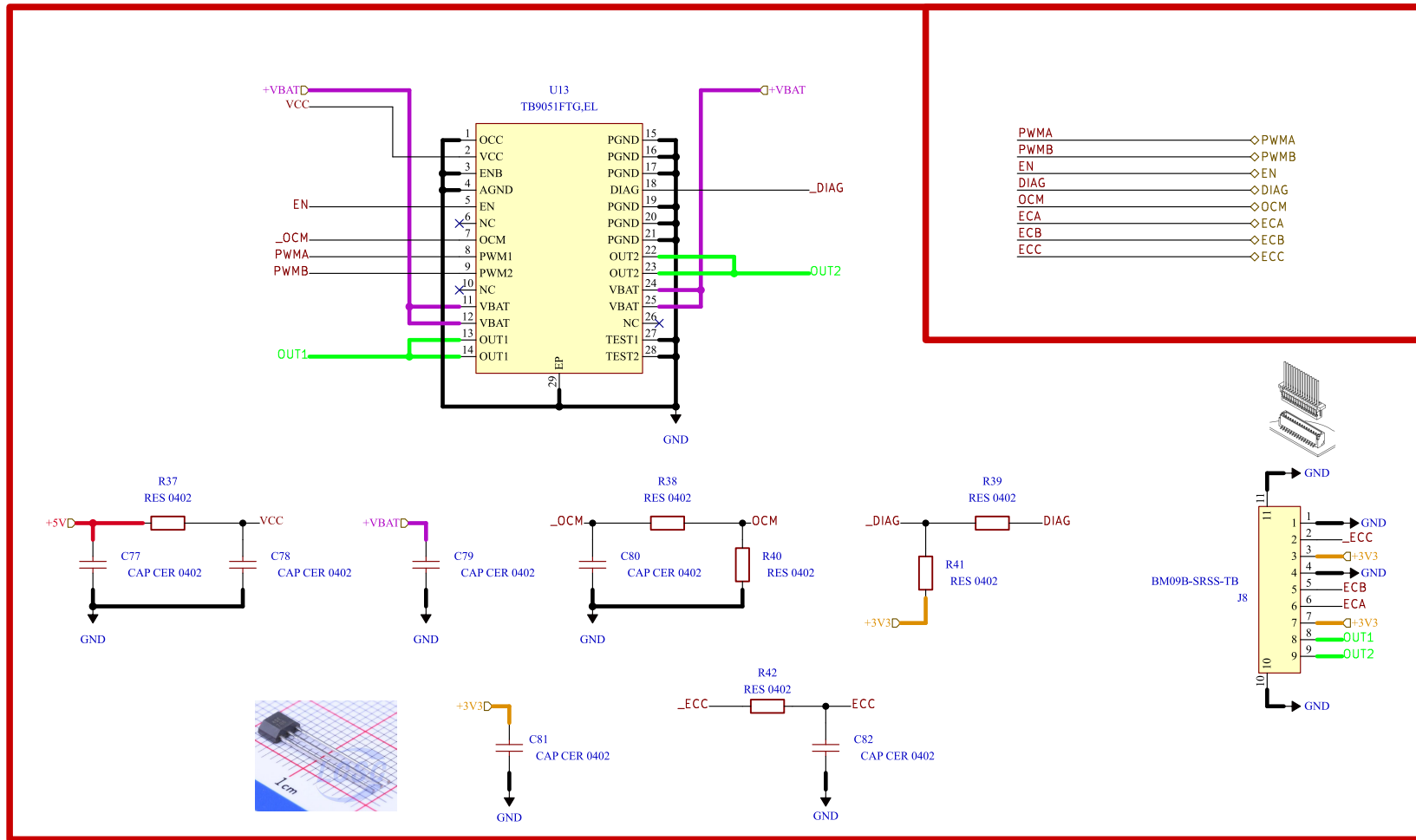
[11] Motor Control 4



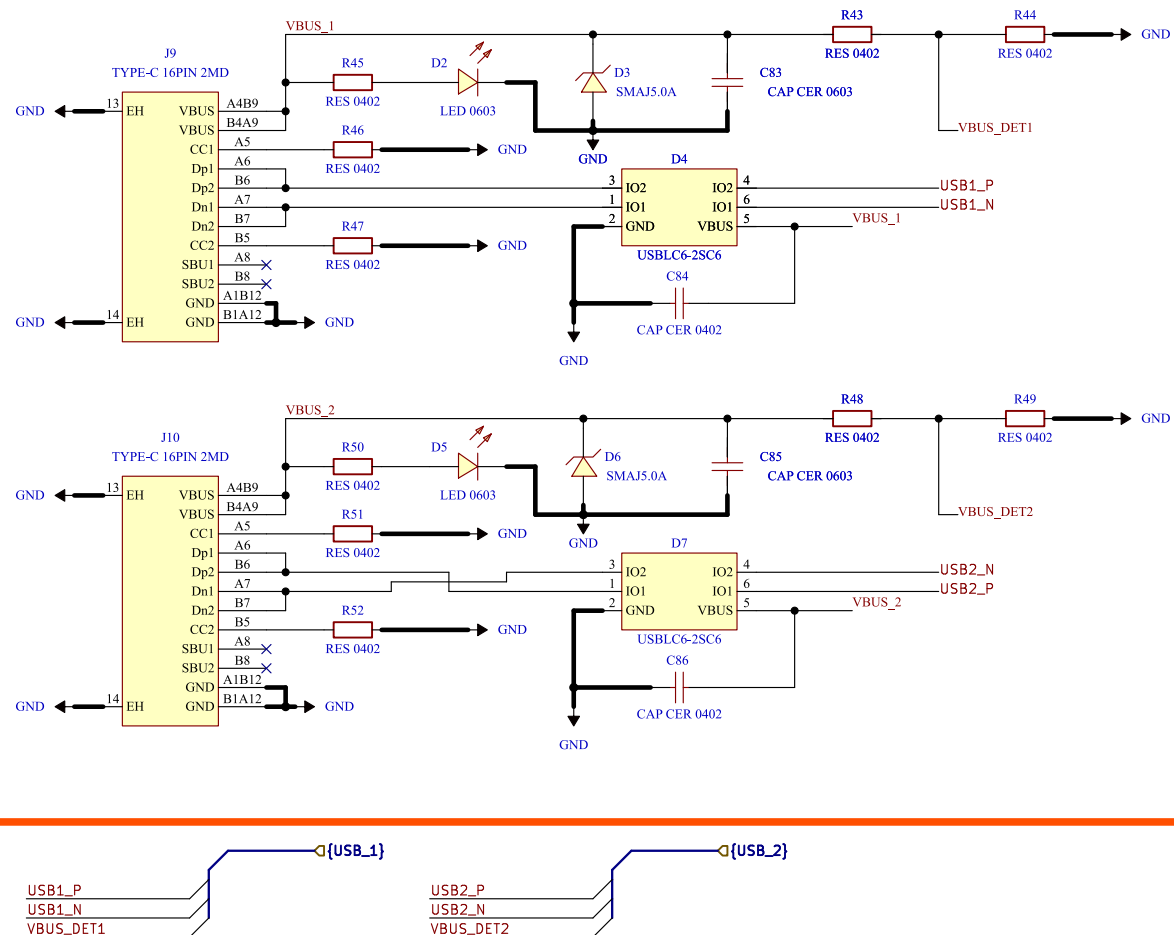
[12] Motor Control 5



[13] Motor Control 6



[14] Interface - USB 2.0



DESIGN NOTE:

[1] VBUS NOT connected to +5V/+3V3. Board is self-powered(LiPo). Sense + LED only. No backfeed to host.

[2] LED direct from VBUS @ 1mA. USB2.0 suspend limit 2.5mA. LED 1mA + divider 33uA = 1.03mA (41%). Do NOT raise to 5mA.

[3] VBUS_DET required - G474 has no dedicated VBUS pin. Self-powered device must not pull up D+ when VBUS=0. FW controls DPPU bit in USB_BCDR. 68k/100k: VBUS 4.40V -> 2.60V > VIH 2.31V. Debounce 100ms.

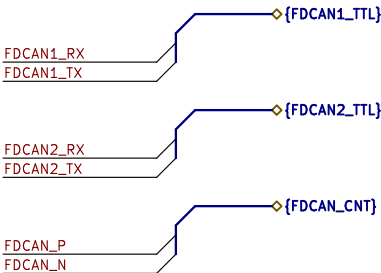
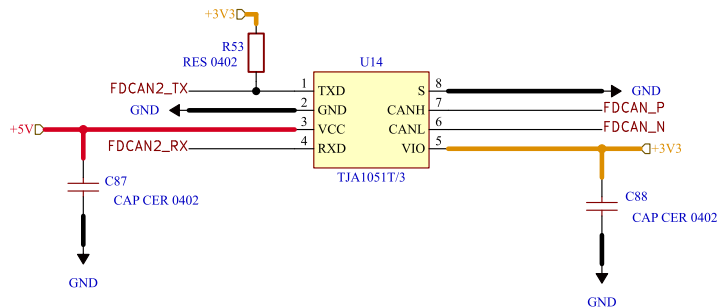
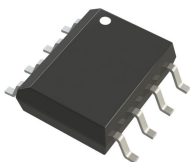
[4] D+/D- direct to PA12/PA11. No 22R series. No external 1k5 pull-up (internal RPUI 900-1500R).

[5] USB clock: PLL-Q 48MHz from HSE 8MHz. ST DFU bootloader uses HSI48+CRS - flashable without crystal.

[6] Both USB ports fully isolated - no shared nets.

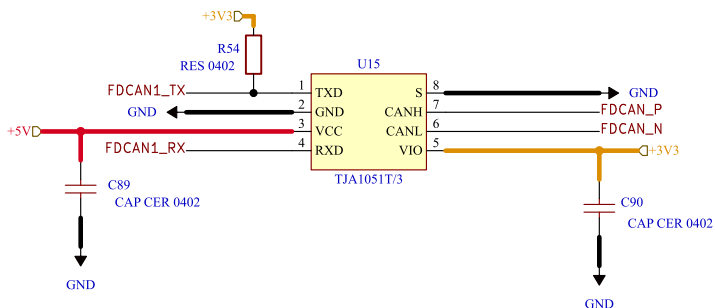
[7] GROUND LOOP WARNING: USB from PC ties host GND to power GND (4.6A peak @20kHz). Use isolator (ADuM3160) or isolated USB-UART when debugging with motors running.

[15] Interface - FD-CAN



DESIGN NOTE:
Silent mode disabled, Pin "S" to ground.

DESIGN NOTE:
120-ohm termination in Interconnects sheet.

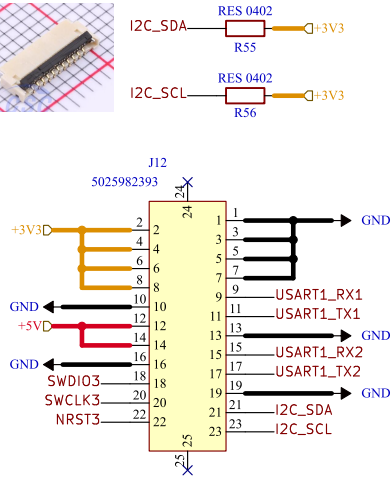
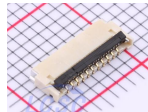


LAYOUT NOTE:
Place close to FD-CAN connector if possible to reduce stub length.

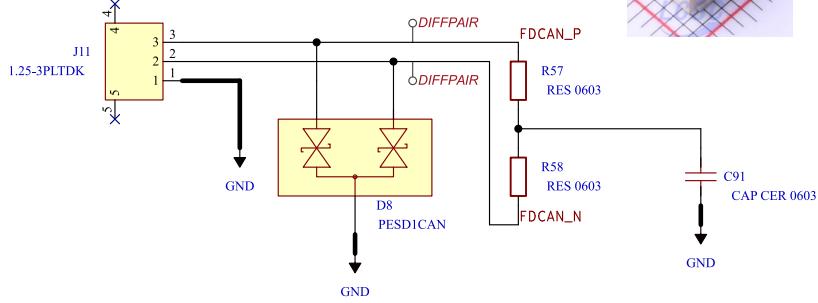
LAYOUT NOTE:
We decided to route it further away to minimize risk of ground lift on the RX/TX lines due to high currents.

Fault Protected FD-CAN Transceiver

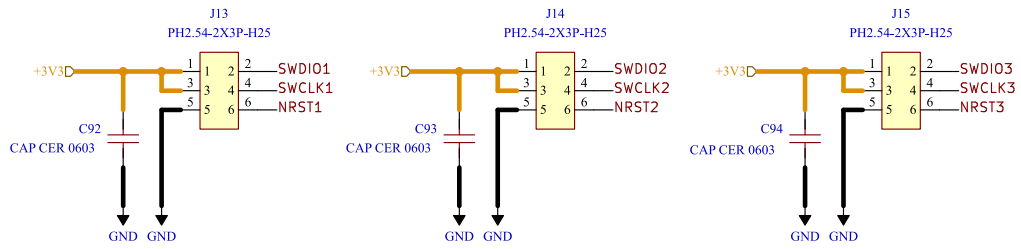
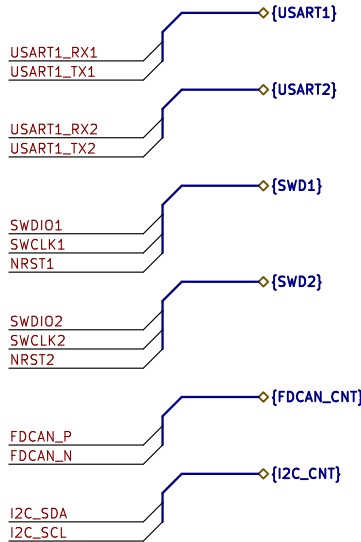
[16] Interface - Interconnects



FFC Connector MCB & MB



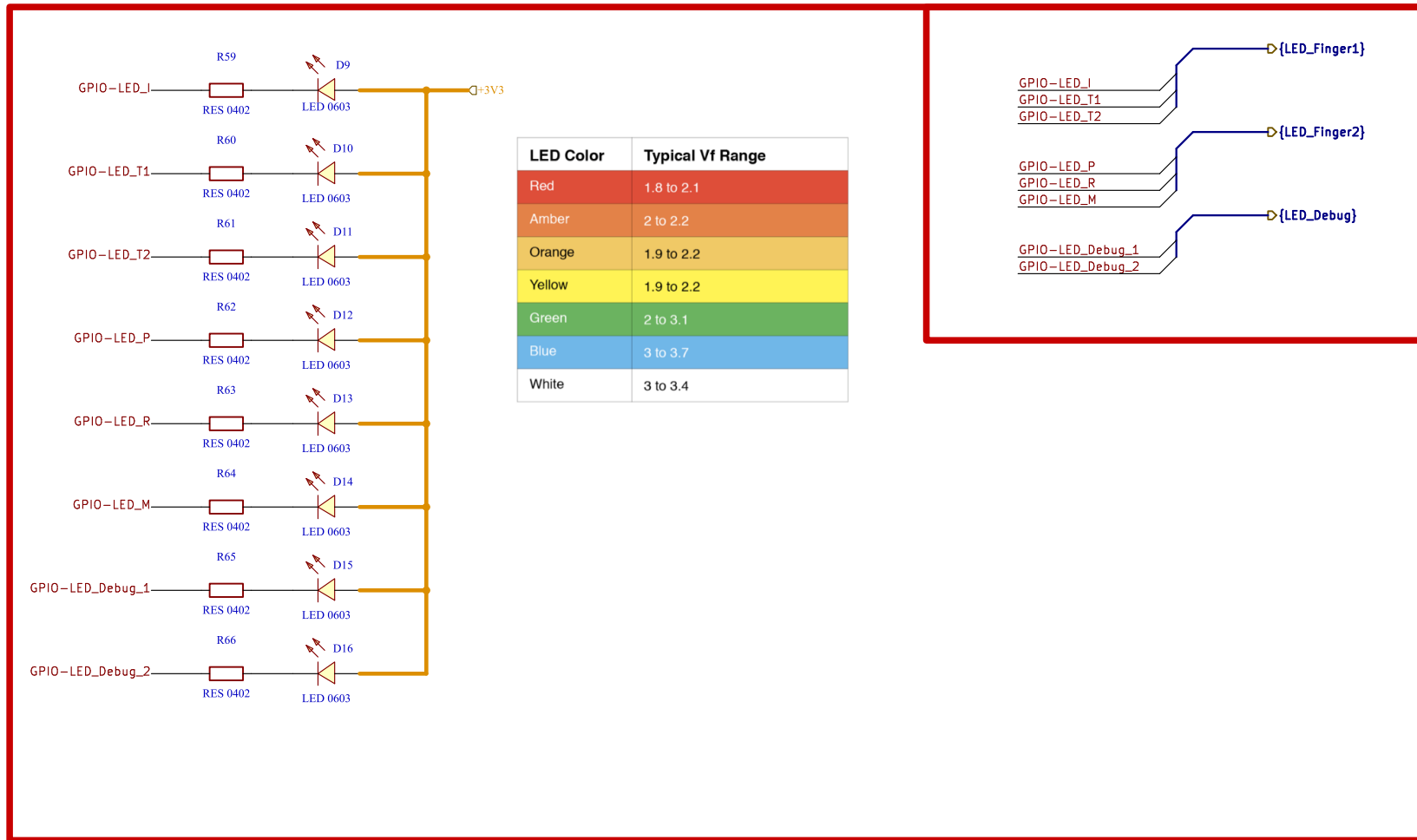
FD-CAN Connector and Termination



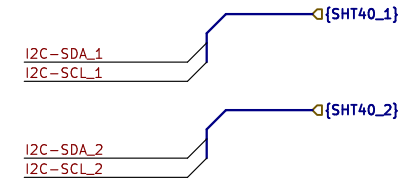
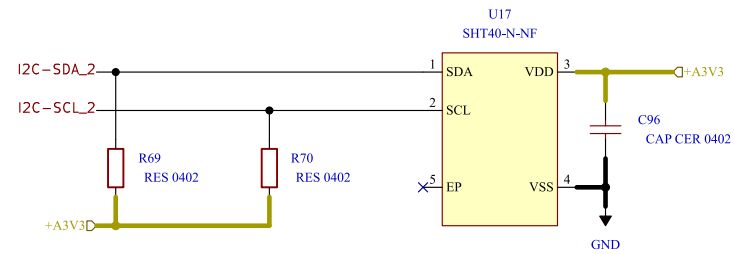
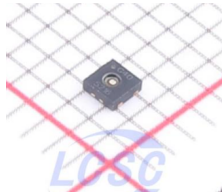
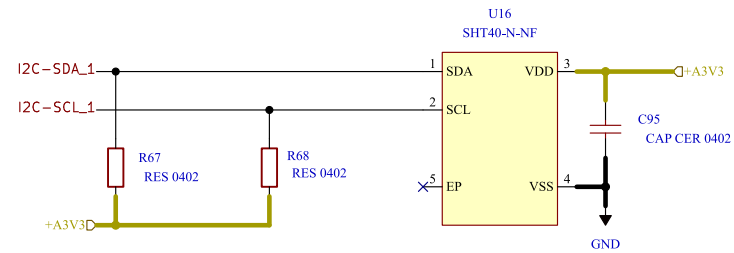
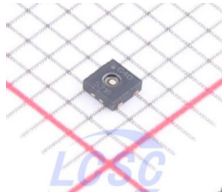
SWD Debug Connector

Reset Button

[17] User - LED Indicators & Buzzer



[18] Sensing - Temperature & Humidity



[19] Sensing - Battery

DESIGN NOTE:
Battery voltage sensing divider for 2S LiPo battery.

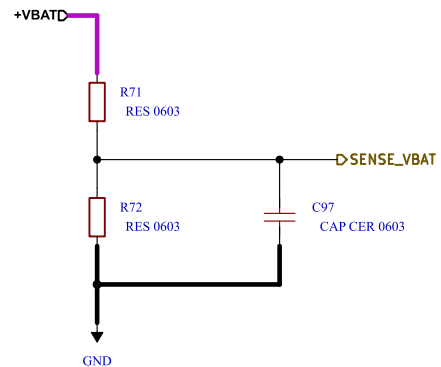
R1 = 100k Ω (Top)
R2 = 47k Ω (Bottom)

Divider Ratio:
 $V_{ADC} = V_{BAT} \times (47 / (100 + 47))$
 $V_{ADC} = V_{BAT} \times 0.3197$

2S LiPo:
Max battery voltage (8.4V): 2.69 V
Nominal voltage (7.4V): 2.37 V
Low battery voltage (6.4V): 2.05 V
Cutoff voltage (6.0V): 1.92 V

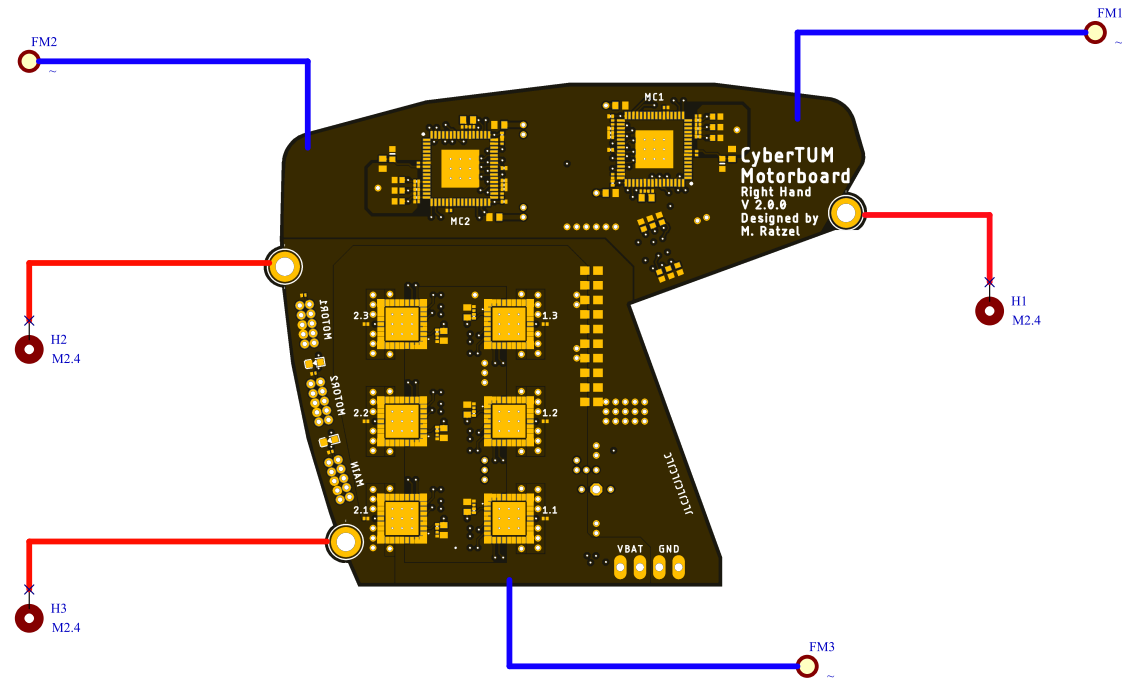
Maximum divider current:
 $I = 8.4V / 147k\Omega = 57 \mu A$

Notes:
- Designed for 3.3V ADC input.
- Add 100nF capacitor at ADC input for noise filtering.
- Use 1% resistors or better.
- Use low temperature coefficient resistors ($\leq 100 \text{ ppm}/^\circ\text{C}$) for improved measurement accuracy.



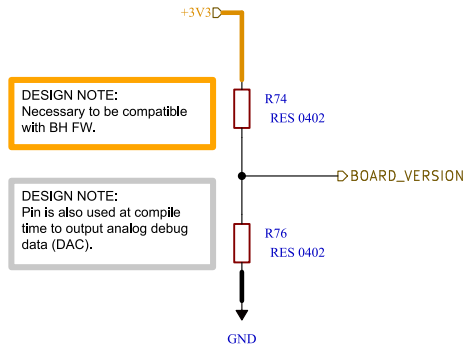
Battery Voltage Sensing

[20] Misc - Holes, Fiducials



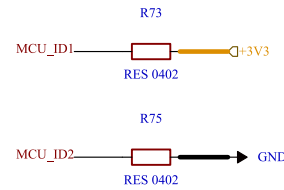
Mounting Holes and Fiducial Markers

[21] Misc - Board Version, Switch

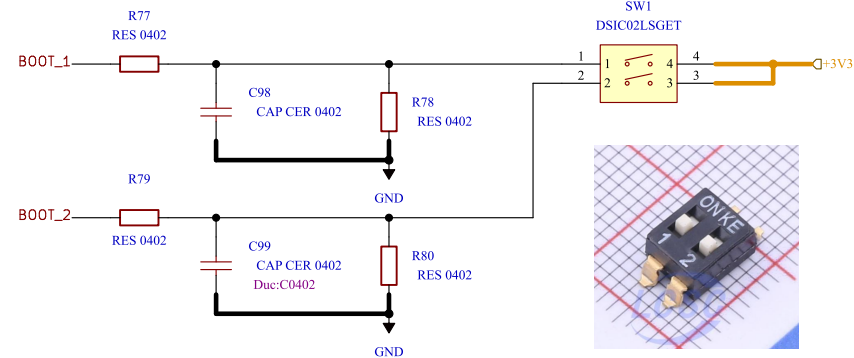
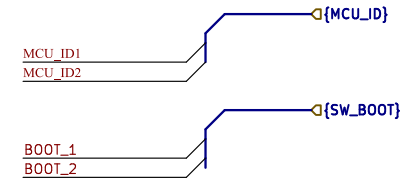


		TOP	BOTTOM
MCB	r1.1, r1.2	×	100k
	r1.3	100k	×
MB	v1.0-1.1	100k	100k

Board Version Resistor Divider



MCU ID



SWITCH BOOT MODE

A

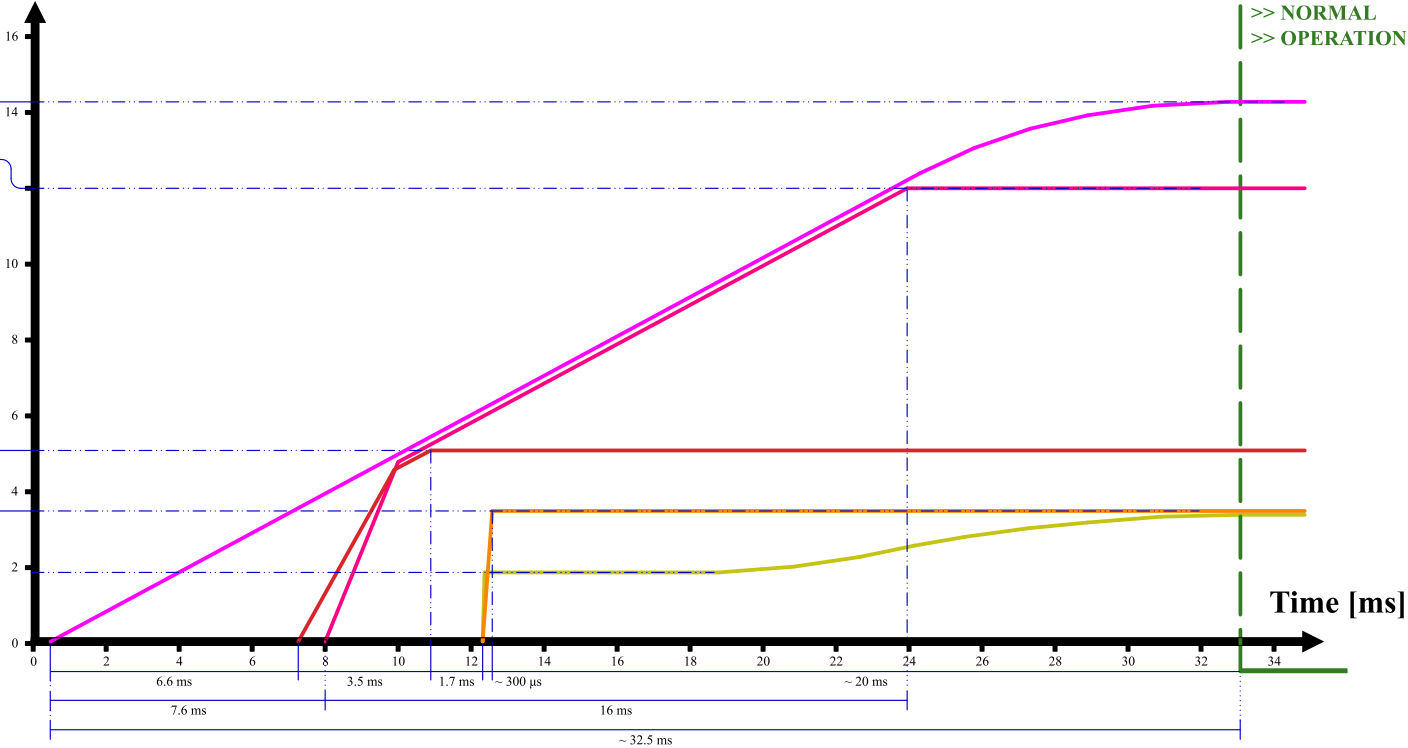
B

C

D

NAME	SOURCE	LEVEL
+V??	??	? - ? ± ??%
+V??	??	? - ? ± ??%
+V??	??	? - ? ± ??%
+V??	??	? - ? ± ??%
+V??	??	? - ? ± ??%

Voltage [V]



	1	2	3	4	5	6
A						
B						
C						
D						

[23]